```
clear;
clc;
close all;
% Design Parameters
Fs = 8000; % Sampling frequency in Hz
Ap = 1; % Passband ripple in dB
As = 40; % Stopband attenuation in dB
% 1. Low-Pass Filter Design
fp_lp = 1000; % LPF Passband frequency (Hz)
fs_lp = 1500; % LPF Stopband frequency (Hz)
wp_lp = fp_lp / (Fs/2);
ws_lp = fs_lp / (Fs/2);
[N_lp, Wn_lp] = buttord(wp_lp, ws_lp, Ap, As);
[b_lp, a_lp] = butter(N_lp, Wn_lp, 'low');
% 2. High-Pass Filter Design
fp_hp = 2500; % HPF Passband frequency (Hz)
fs_hp = 1800; % HPF Stopband frequency (Hz)
wp_hp = fp_hp / (Fs/2);
ws_hp = fs_hp / (Fs/2);
[N_hp, Wn_hp] = buttord(wp_hp, ws_hp, Ap, As);
[b_hp, a_hp] = butter(N_hp, Wn_hp, 'high');
% ==========================================
% DISPLAY NUMERICAL OUTPUTS IN COMMAND WINDOW
% ==========================================
fprintf('=== EXPERIMENT 1: BUTTERWORTH IIR FILTERS ===\n\n');
```

```
=== EXPERIMENT 1: BUTTERWORTH IIR FILTERS ===
```

```
fprintf('--- LOW-PASS FILTER ---\n');
```

```
--- LOW-PASS FILTER ---
```

```
fprintf('Calculated Filter Order (N_lp): %d\n', N_lp);
```

```
Calculated Filter Order (N_lp): 12
```

```
fprintf('3-dB Cutoff Frequency (Wn_lp): %.4f Hz\n', Wn_lp * (Fs/2));
```

```
3-dB Cutoff Frequency (Wn_lp): 1087.8354 Hz
```

```
fprintf('Numerator Coefficients (b_lp):\n');
```

```
Numerator Coefficients (b_lp):
```

```
disp(b_lp);
```

```
    0.0000    0.0000    0.0002    0.0006    0.0013    0.0021    0.0024    0.0021    0.0013    0.0006    0.
```

```
fprintf('Denominator Coefficients (a_lp):\n');
```

```
Denominator Coefficients (a_lp):
```

```
disp(a_lp);
```

```
    1.0000   -5.4622   14.6771  -25.1616   30.3497  -26.9537   17.9862   -9.0534    3.4018   -0.9283    0.
```

```
fprintf('--- HIGH-PASS FILTER ---\n');
```

```
--- HIGH-PASS FILTER ---
```

```
fprintf('Calculated Filter Order (N_hp): %d\n', N_hp);
```

```
Calculated Filter Order (N_hp): 10
```

```
fprintf('3-dB Cutoff Frequency (Wn_hp): %.4f Hz\n', Wn_hp * (Fs/2));
```

```
3-dB Cutoff Frequency (Wn_hp): 2379.7556 Hz
```

```
fprintf('Numerator Coefficients (b_hp):\n');
```

```
Numerator Coefficients (b_hp):
```

```
disp(b_hp);
```

```
    0.0006   -0.0055    0.0248   -0.0662    0.1159   -0.1390    0.1159   -0.0662    0.0248   -0.0055    0.
```

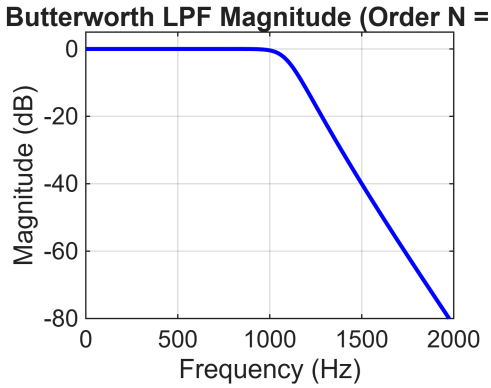
```
fprintf('Denominator Coefficients (a_hp):\n');
```

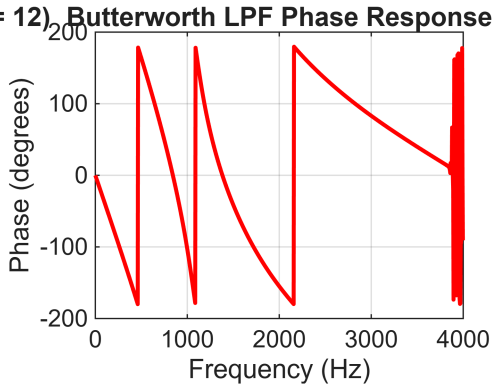
```
Denominator Coefficients (a_hp):
```

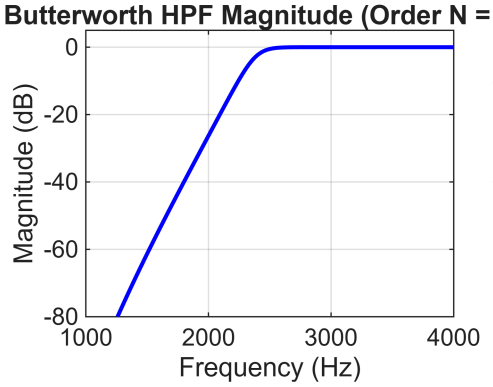
```
disp(a_hp);
```

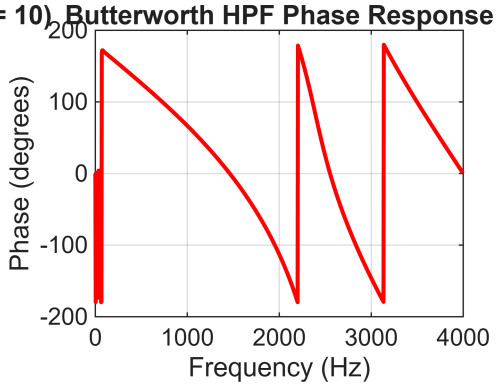
```
    1.0000    1.8915    2.8535    2.6014    1.8704    0.9534    0.3726    0.1030    0.0199    0.0023    0.
```

```
% Frequency Response Calculation
[H_lp, f] = freqz(b_lp, a_lp, 1024, Fs);
[H_hp, ~] = freqz(b_hp, a_hp, 1024, Fs);
% Plotting
figure('Name', 'Butterworth IIR LPF and HPF');
subplot(2,2,1);
plot(f, 20*log10(abs(H_lp)), 'b', 'LineWidth', 1.5); grid on;
title(['Butterworth LPF Magnitude (Order N = ', num2str(N_lp), ')']);
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)'); ylim([-80 5]);
subplot(2,2,2);
plot(f, angle(H_lp)*180/pi, 'r', 'LineWidth', 1.5); grid on;
title('Butterworth LPF Phase Response');
xlabel('Frequency (Hz)'); ylabel('Phase (degrees)');
subplot(2,2,3);
plot(f, 20*log10(abs(H_hp)), 'b', 'LineWidth', 1.5); grid on;
title(['Butterworth HPF Magnitude (Order N = ', num2str(N_hp), ')']);
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)'); ylim([-80 5]);
subplot(2,2,4);
plot(f, angle(H_hp)*180/pi, 'r', 'LineWidth', 1.5); grid on;
title('Butterworth HPF Phase Response');
xlabel('Frequency (Hz)'); ylabel('Phase (degrees)');
```


Butterworth LPF Magnitude (Order N = 12)
Magnitude (dB)
0
-20
-40
-60
-80
0
500
1000
1500
2000
Frequency (Hz)


Butterworth LPF Phase Response
Phase (degrees)
200
100
0
-100
-200
0
1000
2000
3000
4000
Frequency (Hz)


Butterworth HPF Magnitude (Order N = 10)
Magnitude (dB)
0
-20
-40
-60
-80
1000
2000
3000
4000
Frequency (Hz)


Butterworth HPF Phase Response
Phase (degrees)
200
100
0
-100
-200
0
1000
2000
3000
4000
Frequency (Hz)